

Twin Spaces as 2-Categorical and Operadic Objects

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Abstract

In Article I we introduced *twin operations* attached to a group action, and in Article A we developed a deformation cohomology theory for twin spaces via a Hochschild-type complex $C^\bullet(G, \text{End}(E))$. The goal of the present article is to show that twin spaces carry a rich higher-structural content: they naturally form a 2-category, admit a monoidal structure, and can be organized into operads whose algebras encode classical and new symmetry-compatible algebraic structures.

More precisely, we define a 2-category $\mathcal{T}$ whose objects are twin spaces (E, α, G) , whose 1-morphisms are equivariant structure-preserving maps, and whose 2-morphisms are natural transformations compatible with the twin operations. We construct a tensor product making $\mathcal{T}$ into a monoidal 2-category. We then introduce a family of operads $\mathcal{O}_G$ of twin operations associated with a group G , and show that many standard structures (associative G -algebras, G -equivariant module categories, and certain crossed product constructions) appear as algebras over these operads.

Conceptually, this article clarifies the categorical level at which twin spaces live: they are not just sets with operations, but 2-categorical and operadic objects whose deformation theory (Article A) can be viewed as the first step towards a higher, possibly derived, theory of twin structures.

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1 Introduction

Twin operations were introduced in Article I as a new way to attach algebraic structure to a group action: given a set E with a left action of a group G and a binary operation $\alpha : E \times E \rightarrow E$, one considers the family of *twin operations*

$$\alpha_g(x, y) := g^{-1} \cdot \alpha(g \cdot x, g \cdot y), \quad g \in G. \quad (1.1)$$

The set $O = \{\alpha_g : g \in G\}$ of such operations is canonically identified with the homogeneous space $L \backslash G$ (right cosets; see Proposition 1), where $L = \{g \in G : \alpha_g = \alpha\}$ is the linear subgroup. A triple (E, α, G) with this orbit structure was called a *twin space*.

In Article A, we showed that twin spaces admit a deformation cohomology theory, based on a Hochschild-type complex $C^\bullet(G, \mathrm{End}(E))$ whose cohomology $\mathrm{HH}^\bullet(G, \mathrm{End}(E))$ controls G -equivariant deformations and rigidity of the twin structure.

The aim of the present article is to study twin spaces from the *higher-structural* point of view. In particular:

- we construct a 2-category $\mathcal{T}$, with twin spaces as objects, equivariant morphisms as 1-morphisms, and compatible natural transformations as 2-morphisms;
- we equip $\mathcal{T}$ with a tensor product making it a monoidal 2-category in the sense of higher category theory [2, 3, 4];
- for a fixed group G , we organize G -equivariant n -ary twin operations into an operad $\mathcal{O}_G$, in the sense of May and Loday–Vallette [5, 6], and show that various algebraic structures correspond to algebras over $\mathcal{O}_G$;
- we interpret some aspects of the deformation theory of twin spaces (Article A) as living inside this 2-categorical and operadic environment.

Conceptually, the message is that twin spaces form not just a set-theoretic or algebraic construction, but fit naturally into the framework of higher category theory and operads. This clarifies the role of twin spaces as a *foundational* object for subsequent work on derived, homotopical and quantum generalizations.

There are three independent reasons to expect this higher structure, and the article makes each precise. First, the very definition of a twin space produces not one operation but an entire orbit $O \cong L \backslash G$, and an orbit of objects always asks for morphisms relating its points—so a merely 1-categorical packaging is incomplete. Second, the deformation theory of Article A is controlled by a differential graded Lie algebra (its Maurer–Cartan equation), and DGLAs are the infinitesimal shadow of operadic and higher-categorical structure. Third, twin

operations of higher arity compose by substitution exactly as the operations of an operad do. The 2-category, the monoidal structure, and the operad constructed below are the precise homes for these three observations.

The article is organized as follows. In Section 2, we define the 2-category $\mathcal{T}$ of twin spaces. In Section 3, we construct a monoidal structure on $\mathcal{T}$. In Section 4, we introduce the operads of twin operations $\mathcal{O}_G$ and describe their algebras. In Section 6, we discuss functoriality and 2-functors relating $\mathcal{T}$ to more classical categories. In Section 7, we sketch how the deformation cohomology of Article A can be seen from a 2-categorical/operadic point of view. Finally, in Section 8 we illustrate the constructions on concrete examples.

2 Twin Spaces as Objects of a 2-Category

We begin by recalling the basic definition of a twin space and then organize twin spaces into a 2-category.

2.1 Twin spaces

Let G be a group acting on a set E from the left. Given a binary operation $\alpha : E \times E \rightarrow E$, we define the family of twin operations by (1.1). Write again

$$O = \{\alpha_g : g \in G\}.$$

Proposition 1 (Orbit identification). *The map $g \mapsto \alpha_g$ has fibres the right cosets of $L = \{g \in G : \alpha_g = \alpha\}$: indeed $\alpha_{g_1} = \alpha_{g_2}$ iff $g_1 g_2^{-1} \in L$, i.e. iff $L g_1 = L g_2$. Hence $O \cong L \backslash G$ (right cosets); when $L \trianglelefteq G$ this coincides with G/L .*

Definition 2 (Twin space). A *twin space* is a triple (E, α, G) , where:

- E is a set (or object in a fixed ambient category, e.g. a vector space),
- G is a group acting on E ,
- $\alpha : E \times E \rightarrow E$ is a binary operation such that the orbit O of (1.1) is well-defined and carries the induced G -action.

The associated *linear subgroup* is $L := \{g \in G : \alpha_g = \alpha\}$.

In many settings, one requires α to satisfy extra identities (associativity, commutativity, etc.), but for the 2-categorical structure the basic notion suffices.

2.2 1-morphisms: equivariant structure-preserving maps

We want morphisms that preserve both the G -action and the twin operation structure.

Definition 3 (1-morphisms of twin spaces). Let (E, α, G) and (E', α', G') be twin spaces. A *1-morphism*

$$F : (E, \alpha, G) \longrightarrow (E', \alpha', G')$$

consists of:

- a group homomorphism $\phi : G \rightarrow G'$,
- a map of sets (or linear map) $f : E \rightarrow E'$,

such that:

(a) f is ϕ -equivariant:

$$f(g \cdot x) = \phi(g) \cdot f(x), \quad \forall g \in G, x \in E;$$

(b) f is compatible with the operations:

$$f(\alpha(x, y)) = \alpha'(f(x), f(y)), \quad \forall x, y \in E.$$

Remark 4. Condition (b) can be strengthened if needed to require compatibility with *all* twin operations α_g and $\alpha'_{\phi(g)}$, but this already follows from (a) and the definition of twin operations:

$$f(\alpha_g(x, y)) = f(g^{-1} \cdot \alpha(g \cdot x, g \cdot y)) = \phi(g)^{-1} \cdot f(\alpha(g \cdot x, g \cdot y)) = \phi(g)^{-1} \cdot \alpha'(f(g \cdot x), f(g \cdot y)),$$

and $f(g \cdot x) = \phi(g) \cdot f(x)$ implies

$$f(\alpha_g(x, y)) = (\alpha')_{\phi(g)}(f(x), f(y)).$$

Example 5 (The determinant as a 1-morphism). Let (E, α, G) be the matrix twin space $E = M_n(k)$, $\alpha(X, Y) = XY$, with $G = GL_n(k)$ acting by conjugation, and let $(E', \alpha', G') = (k, \cdot, \{1\})$ be the multiplicative twin space on the ground field with trivial group. The pair $F = (\det, \phi)$, with $f = \det : M_n(k) \rightarrow k$ and $\phi : GL_n(k) \rightarrow \{1\}$ the trivial homomorphism, is a 1-morphism: equivariance holds because $\det(gXg^{-1}) = \det X = \phi(g) \cdot \det X$, and operation-compatibility is the multiplicativity $\det(XY) = \det X \det Y$. Thus the determinant is not merely a function but a morphism of twin spaces, collapsing the conjugation-invariant matrix multiplication onto ordinary multiplication of scalars.

2.3 2-morphisms: compatible natural transformations

We now define 2-morphisms between 1-morphisms.

Definition 6 (2-morphisms of twin spaces). Let $F = (f, \phi)$ and $F' = (f', \phi')$ be 1-morphisms from (E, α, G) to (E', α', G') . A *2-morphism*

$$\eta : F \Rightarrow F'$$

is a family of maps

$$\eta_x : f(x) \rightarrow f'(x), \quad x \in E,$$

in the ambient category (typically $\text{id}_{f(x)}$ if one stays within sets or vector spaces), such that:

(a) η is G -equivariant:

$$\eta_{g \cdot x} = \phi(g) \cdot \eta_x, \quad \forall g \in G, x \in E,$$

which is well-typed because $f(g \cdot x) = \phi(g) \cdot f(x)$ and $f'(g \cdot x) = \phi'(g) \cdot f'(x)$, so that the translate $\phi(g) \cdot \eta_x$ is again a morphism $f(g \cdot x) \rightarrow f'(g \cdot x)$;

(b) η is compatible with the operations:

$$\eta_{\alpha(x, y)} = \alpha'(\eta_x, \eta_y),$$

Remark 7. In the purely set-theoretic case, the above notion collapses to the identity transformation (i.e. 2-morphisms are trivial). The main interest is in enriched contexts, e.g. when E is a category or a module object, in which case η_x is a genuine morphism in that category. In that enriched setting the two conditions read transparently: equivariance says the component

at $g \cdot x$ is the $\phi(g)$ -translate of the component at x , while operation-compatibility says that evaluating η at $\alpha(x, y)$ agrees with applying the operation α' , viewed as a bifunctor, to the pair (η_x, η_y) . When the ambient category is **Set** or **Vect** the only available components force $f = f'$ and $\eta = \text{id}$, which is exactly why the genuinely 2-categorical content surfaces only after enrichment (for instance in **Cat**, where a 2-morphism is a G -equivariant natural transformation between equivariant functors). For clarity, we adopt the 2-categorical language even though in some basic examples the 2-morphisms are only identities.

2.4 The 2-category $\mathcal{T}$

Theorem 8. *Twin spaces, 1-morphisms and 2-morphisms as above form a (strict) 2-category $\mathcal{T}$.*

Proof. The composition of 1-morphisms is defined componentwise: if $F = (f, \phi) : (E, \alpha, G) \rightarrow (E', \alpha', G')$ and $F' = (f', \phi') : (E', \alpha', G') \rightarrow (E'', \alpha'', G'')$, then

$$F' \circ F = (f' \circ f, \phi' \circ \phi).$$

Equivariance and compatibility with operations are preserved by composition. The identity 1-morphism is $(\text{id}_E, \text{id}_G)$.

The vertical and horizontal compositions of 2-morphisms are defined in the usual way for natural transformations, and the coherence axioms of a 2-category are satisfied by standard arguments (see e.g. [2, 3]). The middle-four interchange law holds because it holds in the ambient enriched category, and both the equivariance condition $\eta_{g \cdot x} = \phi(g) \cdot \eta_x$ and the operation-compatibility condition are visibly closed under vertical and horizontal composition. Thus we obtain a strict 2-category. $\square$

3 Monoidal Structure on the 2-Category $\mathcal{T}$

We now equip $\mathcal{T}$ with a tensor product, making it into a monoidal 2-category.

3.1 Tensor product of twin spaces

Let (E, α, G) and (E', α', G') be twin spaces.

Definition 9 (Tensor product of twin spaces). The *tensor product* of (E, α, G) and (E', α', G') is the twin space

$$(E \otimes E', \alpha \otimes \alpha', G \times G'),$$

defined as follows:

- the underlying object is $E \otimes E'$ (e.g. the cartesian product $E \times E'$ for sets, the tensor product for vector spaces);
- the group $G \times G'$ acts componentwise:

$$(g, g') \cdot (x, x') := (g \cdot x, g' \cdot x');$$

- the operation $\alpha \otimes \alpha'$ is given by

$$(\alpha \otimes \alpha')((x_1, x'_1), (x_2, x'_2)) := (\alpha(x_1, x_2), \alpha'(x'_1, x'_2)).$$

Proposition 10. *The triple $(E \otimes E', \alpha \otimes \alpha', G \times G')$ is a twin space, and twin operations satisfy*

$$(\alpha \otimes \alpha')_{(g, g')} = \alpha_g \otimes \alpha'_{g'}.$$

Proof. One checks directly that

$$(\alpha \otimes \alpha')_{(g, g')}((x_1, x'_1), (x_2, x'_2)) = (g, g')^{-1} \cdot (\alpha \otimes \alpha')((g, g') \cdot (x_1, x'_1), (g, g') \cdot (x_2, x'_2))$$

unwinds, using the componentwise action and the definition of $\alpha \otimes \alpha'$, as

$$(g^{-1}, g'^{-1}) \cdot (\alpha(g \cdot x_1, g \cdot x_2), \alpha'(g' \cdot x'_1, g' \cdot x'_2)) = (g^{-1} \cdot \alpha(g \cdot x_1, g \cdot x_2), g'^{-1} \cdot \alpha'(g' \cdot x'_1, g' \cdot x'_2)) = (\alpha_g(x_1, x_2), \alpha'_{g'}(x'_1, x'_2))$$

which is precisely $(\alpha_g \otimes \alpha'_{g'})((x_1, x'_1), (x_2, x'_2))$. That the triple is a twin space is then immediate: the orbit of $\alpha \otimes \alpha'$ is the product of the two orbits. $\square$

Remark 11 (Linear subgroup of a tensor product). The linear subgroup of the tensor product is the product of the linear subgroups, $L_{E \otimes E'} = L \times L'$, because $(\alpha \otimes \alpha')_{(g, g')} = \alpha \otimes \alpha'$ holds iff $\alpha_g = \alpha$ and $\alpha'_{g'} = \alpha'$. Hence the twin orbit of a tensor product factors as a product of orbits,

$$(L \times L') \backslash (G \times G') \cong (L \backslash G) \times (L' \backslash G'),$$

so the twin-orbit 2-functor of Section 6 is itself monoidal.

3.2 Tensor product of morphisms and monoidal axioms

Definition 12. Given 1-morphisms $F = (f, \phi) : (E, \alpha, G) \rightarrow (E_1, \alpha_1, G_1)$ and $F' = (f', \phi') : (E', \alpha', G') \rightarrow (E'_1, \alpha'_1, G'_1)$, we define

$$F \otimes F' := (f \otimes f', \phi \times \phi'),$$

where

$$(f \otimes f')(x, x') := (f(x), f'(x')), \quad (\phi \times \phi')(g, g') := (\phi(g), \phi'(g')).$$

Tensor product of 2-morphisms is defined pointwise.

Theorem 13. *Equipped with the tensor product above, the 2-category $\mathcal{T}$ becomes a monoidal 2-category.*

Proof. Associativity and unit constraints are inherited from the ambient category (e.g. Sets or Vect), and the coherence axioms of a monoidal 2-category follow from standard arguments as in [2, 3]. $\square$

Remark 14. In many concrete settings (e.g. vector spaces over a fixed field), the monoidal structure is in fact strict or can be strictified without loss of generality, so that associativity and unit isomorphisms can be taken to be identities.

4 Operads of Twin Operations

We now introduce a family of operads whose operations are twin operations compatible with a given group action.

4.1 The underlying collections

Fix a group G and a G -set (or G -object) E .

Definition 15 (Twin n -ary operations). For $n \geq 1$, let

$$\mathcal{O}_G(n) := \{\alpha : E^n \rightarrow E \mid \alpha \text{ is } G\text{-equivariant in the sense of twin operations}\}.$$

More precisely, $\alpha \in \mathcal{O}_G(n)$ if there exists a family

$$\alpha_g : E^n \rightarrow E, \quad g \in G,$$

such that

$$\alpha_g(x_1, \dots, x_n) = g^{-1} \cdot \alpha(g \cdot x_1, \dots, g \cdot x_n),$$

and the map $g \mapsto \alpha_g$ is compatible with the group action (in the same spirit as in the binary case).

We think of $\mathcal{O}_G(n)$ as the space of n -ary twin operations on E .

Remark 16 ($\mathcal{O}_G$ is the endomorphism operad internal to G -sets). The existence of a transport family $g \mapsto \alpha_g$ is automatic: *every* map $\alpha : E^n \rightarrow E$ admits one, namely $\alpha_g(\mathbf{x}) = g^{-1} \cdot \alpha(g \cdot \mathbf{x})$. Thus, set-theoretically, $\mathcal{O}_G(n) = \text{Maps}(E^n, E)$ is the full endomorphism operad $\text{End}_E(n)$; the twin viewpoint adds not more operations but more *structure*, namely the G -action $g \cdot \alpha := \alpha_{g^{-1}}$ on each $\text{End}_E(n)$, compatible with substitution. In one phrase, $\mathcal{O}_G$ is the endomorphism operad *internal to the category of G -sets*, and the linear subgroup L of a chosen α is simply its stabiliser under this action.

4.2 Operadic composition

Let $\alpha \in \mathcal{O}_G(k)$ and $\beta_1 \in \mathcal{O}_G(n_1), \dots, \beta_k \in \mathcal{O}_G(n_k)$. We define their *substitution*:

$$\gamma = \alpha \circ (\beta_1, \dots, \beta_k) \in \mathcal{O}_G(n_1 + \dots + n_k)$$

by

$$\gamma(x_{1,1}, \dots, x_{1,n_1}; \dots; x_{k,1}, \dots, x_{k,n_k}) = \alpha(\beta_1(x_{1,1}, \dots, x_{1,n_1}), \dots, \beta_k(x_{k,1}, \dots, x_{k,n_k})).$$

Proposition 17. *The substitution $\circ$ above defines a well-defined operation*

$$\mathcal{O}_G(k) \times \mathcal{O}_G(n_1) \times \dots \times \mathcal{O}_G(n_k) \rightarrow \mathcal{O}_G(n_1 + \dots + n_k),$$

compatible with the G -equivariance structure of twin operations.

Proof. We must show that if α and each β_i admit twin families (α_g) and $(\beta_{i,g})$, then γ admits a twin family (γ_g) given by the same transport rule, i.e.

$$\gamma_g(\mathbf{x}) = g^{-1} \cdot \gamma(g \cdot \mathbf{x}),$$

with $\mathbf{x}$ denoting the whole tuple of variables. Concretely, transport distributes over substitution. Using $\alpha_g(y_1, \dots, y_k) = g^{-1} \cdot \alpha(g \cdot y_1, \dots, g \cdot y_k)$ and $\beta_{i,g}(\mathbf{x}_i) = g^{-1} \cdot \beta_i(g \cdot \mathbf{x}_i)$, we compute

$$\alpha_g(\beta_{1,g}(\mathbf{x}_1), \dots, \beta_{k,g}(\mathbf{x}_k)) = g^{-1} \cdot \alpha(g \cdot \beta_{1,g}(\mathbf{x}_1), \dots) = g^{-1} \cdot \alpha(\beta_1(g \cdot \mathbf{x}_1), \dots, \beta_k(g \cdot \mathbf{x}_k)) = g^{-1} \cdot \gamma(g \cdot \mathbf{x}) = \gamma_g(\mathbf{x}),$$

using $g \cdot \beta_{i,g}(\mathbf{x}_i) = \beta_i(g \cdot \mathbf{x}_i)$ in the middle step. Hence $\gamma_g = \alpha_g \circ (\beta_{1,g}, \dots, \beta_{k,g})$: the transport $\alpha \mapsto \alpha_g$ is an operad endomorphism of End_E , so a substitution of twin operations is again a twin operation. $\square$

4.3 Symmetries and the symmetric operad structure

If G acts on E and the symmetric group S_n acts on E^n by permuting factors, we can also consider the interaction of twins with these symmetric group actions.

Definition 18. For $\sigma \in S_n$ and $\alpha \in \mathcal{O}_G(n)$, define

$$(\alpha \cdot \sigma)(x_1, \dots, x_n) := \alpha(x_{\sigma^{-1}(1)}, \dots, x_{\sigma^{-1}(n)}).$$

Proposition 19. *The collection $(\mathcal{O}_G(n))_{n \geq 1}$ with the substitution maps and the S_n -actions defined above forms a (symmetric) operad in the sense of [5, 6].*

Proof. The axioms of a symmetric operad (associativity of substitution, unit, equivariance with respect to symmetric groups) are checked as in the standard example of the endomorphism operad of E , with the further observation that the twin structure is preserved because it is defined by transport along the group action, which commutes with the S_n -action. $\square$

Definition 20 (Twin operad). We call $\mathcal{O}_G$ the *twin operad* associated with the G -object E .

5 Algebras over Twin Operads and Classical Structures

Example 21 (Small cases of the twin operad). When $G = \{e\}$ the action is trivial and $\mathcal{O}_{\{e\}} = \text{End}_E$ is the ordinary endomorphism operad; a chosen associative α then determines an operad morphism $\text{Assoc} \rightarrow \text{End}_E$, as used below. For finite E with $|E| = m$ each arity is finite, $|\mathcal{O}_G(n)| \leq m^{m^n}$; for instance with $E = \mathbb{Z}/2$ there are $m^{m^2} = 2^4 = 16$ binary operations, among which the unital associative commutative ones realise E as a monoid. The twin operad refines this finite set by recording how G permutes the operations through transport.

We now explain how familiar objects appear as algebras over $\mathcal{O}_G$.

5.1 Twin operad algebras

Definition 22. Let $\mathcal{O}$ be an operad in a symmetric monoidal category $\mathcal{C}$. An $\mathcal{O}$ -*algebra* is an object $X \in \mathcal{C}$ together with structure maps

$$\mathcal{O}(n) \otimes X^{\otimes n} \rightarrow X$$

satisfying the usual associativity, unit, and equivariance axioms.

Definition 23 (Algebras over the twin operad). An $\mathcal{O}_G$ -*algebra* is an object X of the ambient category equipped with a morphism of operads

$$\mathcal{O}_G \rightarrow \text{End}_X,$$

where $\text{End}_X(n) = \text{Hom}(X^{\otimes n}, X)$ is the endomorphism operad of X .

Remark 24. Concretely, an $\mathcal{O}_G$ -algebra is a G -object X endowed with twin operations acting on X in a way compatible with operadic composition.

5.2 Associative G -equivariant algebras

Let G act by automorphisms on an associative algebra A .

Proposition 25. *If the action of G preserves the associative multiplication of A , then A is naturally an algebra over a quotient of $\mathcal{O}_G$ corresponding to the associative operad Assoc .*

Proof. The associative operad **Assoc** is generated by a single binary operation μ modulo associativity, with $\text{Assoc}(n) \cong k[S_n]$ the $n!$ orderings of the inputs. That G preserves the multiplication of A means μ lies in the L -fixed part of $\mathcal{O}_G(2)$, i.e. it is a genuine G -equivariant (twin) operation. A G -equivariant associative algebra structure on A can be encoded as a morphism of operads

$$\text{Assoc} \rightarrow \mathcal{O}_G \rightarrow \text{End}_A,$$

where the first arrow picks out the twin lift of the associative operation, and the second arrow evaluates it on A . The G -equivariance ensures compatibility with the twin structure. $\square$

5.3 Other structures

Similarly, by choosing appropriate quotients or sub-operads of $\mathcal{O}_G$, one can encode:

- commutative G -equivariant algebras via a commutative quotient of $\mathcal{O}_G$;
- Lie-type twin structures via an operad obtained from $\mathcal{O}_G$ by imposing antisymmetry and Jacobi-type identities;
- module structures and crossed products via colored or multi-sorted versions of $\mathcal{O}_G$.

We do not pursue these constructions further here, as they require a case-by-case analysis depending on the type of algebraic identities involved.

6 Functoriality and 2-Functors

The 2-category $\mathcal{T}$ comes with several natural 2-functors to more classical categories.

6.1 The forgetful 2-functor to G -objects

Definition 26. Define a 2-functor

$$U : \mathcal{T} \rightarrow \mathbf{GObj},$$

where $\mathbf{GObj}$ is the 2-category of pairs (E, G) with group actions, 1-morphisms given by equivariant maps, and 2-morphisms given by natural transformations (when enriched). On objects, $U(E, \alpha, G) = (E, G)$; on morphisms, $U(f, \phi) = (f, \phi)$; and on 2-morphisms, $U(\eta) = \eta$.

Proposition 27. U is a strict 2-functor that is faithful on 1-morphisms and 2-morphisms.

Proof. The functor U simply forgets the twin operation α ; all coherence data for a strict 2-functor are inherited from the structure of $\mathcal{T}$ and $\mathbf{GObj}$. Faithfulness follows from the fact that different 1- or 2-morphisms differ already at the level of the underlying G -objects. $\square$

6.2 The twin-orbit 2-functor

Definition 28. Define a 2-functor

$$\mathcal{O} : \mathcal{T} \rightarrow \mathbf{HomogeneousSpaces},$$

sending (E, α, G) to the homogeneous space $L \backslash G$, and a 1-morphism $F = (f, \phi)$ to the induced map of homogeneous spaces

$$L \backslash G \rightarrow L' \backslash G', \quad Lg \mapsto L' \phi(g).$$

Proposition 29. $\mathcal{O}$ is a well-defined 2-functor on the sub-2-category $\mathcal{T}^{\text{epi}}$ whose 1-morphisms have surjective underlying map f .

Proof. Well-definedness on objects is Proposition 1. For 1-morphisms one must check $\phi(L) \subseteq L'$. Let $g \in L$, so $\alpha(g \cdot x, g \cdot y) = g \cdot \alpha(x, y)$. For $x' = f(x)$, $y' = f(y)$ in the image of f ,

$$\alpha'(\phi(g) \cdot x', \phi(g) \cdot y') = \alpha'(f(g \cdot x), f(g \cdot y)) = f(\alpha(g \cdot x, g \cdot y)) = f(g \cdot \alpha(x, y)) = \phi(g) \cdot \alpha'(x', y').$$

Since f is surjective this holds for all $x', y' \in E'$, hence $\phi(g) \in L'$ and $\phi(L) \subseteq L'$; the induced coset map $Lg \mapsto L'\phi(g)$ is therefore well-defined. (On a non-surjective image the relation $\phi(g) \in L'$ may fail, which is why the hypothesis is needed.) On 2-morphisms, the effect is trivial at the level of homogeneous spaces, so we obtain a strict 2-functor. $\square$

Remark 30. The 2-functor $\mathcal{O}$ encodes the geometric content of the twin structure: it ignores the detailed algebraic operation and remembers only the orbit $L \backslash G$. This reflects the central role of this homogeneous space in the deformation theory of twin spaces.

7 Interaction with Deformation Cohomology

We briefly explain how the deformation cohomology of Article A can be viewed from this 2-categorical and operadic perspective.

7.1 From cochains to 2-endomorphisms

In Article A, the deformation complex

$$C^\bullet(G, \text{End}(E))$$

was constructed as a Hochschild-type complex controlling deformations of (E, α, G) .

Remark 31. At a conceptual level, $C^n(G, \text{End}(E))$ can be thought of as n -fold higher endomorphisms of the identity 1-morphism on (E, α, G) in the 2-category $\mathcal{T}$, enriched appropriately. The differential δ encodes the failure of these higher endomorphisms to be compatible under composition, and the cohomology $\text{HH}^\bullet$ measures higher deformations and obstructions to strictness. This viewpoint parallels the interpretation of Hochschild cohomology as self-extensions in the derived category of bimodules [7].

7.2 Operadic interpretation of HH^1 and HH^2

The twin operad $\mathcal{O}_G$ controls the allowed types of operations on E . A deformation of the twin structure can be seen as a deformation of the operad morphism

$$\mathcal{O}_G \rightarrow \text{End}_E.$$

Proposition 32 (Heuristic). *Infinitesimal deformations parametrized by $\text{HH}^1(G, \text{End}(E))$ correspond to infinitesimal perturbations of the operad morphism $\mathcal{O}_G \rightarrow \text{End}_E$, and obstructions in $\text{HH}^2(G, \text{End}(E))$ correspond to failures of extending such perturbations to higher order.*

Idea of proof. Operadic deformation theory (as in the case of associative algebras) can be formulated in terms of Maurer–Cartan elements in suitable differential graded Lie algebras built from the endomorphism operad. In the twin setting, the group action intertwines with the operadic structure, and the deformation complex acquires the form $C^\bullet(G, \text{End}(E))$. The compatibility between operadic and group-theoretic structures yields the claimed identification, though a full proof would require setting up the operadic deformation complex explicitly, which we leave for future work. $\square$

8 Examples

We conclude with a few examples illustrating the constructions above.

Example 33 (Matrix twin spaces). Let $E = M_n(k)$ with the conjugation action of $G = GL_n(k)$, and let $\alpha(X, Y) = XY$ be matrix multiplication. Then (E, α, G) is a twin space (as in Article A), and 1-morphisms in $\mathcal{T}$ from this object to itself include inner automorphisms of $M_n(k)$ and certain outer automorphisms induced by group automorphisms of $GL_n(k)$.

The twin operad $\mathcal{O}_G$ on E includes all G -equivariant n -ary multilinear operations on $M_n(k)$. Algebras over suitable quotients of $\mathcal{O}_G$ recover associative G -equivariant algebra structures.

Example 34 (Compact Lie group actions on manifolds). Let G be a compact connected Lie group acting smoothly on a manifold E , and let α be a smooth G -equivariant operation (e.g. a pointwise product of functions, or a convolution-type operation). Then (E, α, G) is a twin space in the smooth category, yielding an object of $\mathcal{T}$. The monoidal structure of Section 3 allows us to construct twin spaces on products of manifolds and group actions.

Example 35 (Discretized twin operads). If E is a finite G -set, then $\mathcal{O}_G(n)$ is a finite set of G -equivariant n -ary operations. The twin operad $\mathcal{O}_G$ is thus a finite operad in the category of sets. Algebras over $\mathcal{O}_G$ in the category of sets or vector spaces give discrete or linear models of twin structures, suitable for combinatorial or computational applications.

9 Conclusion and Outlook

In this article we have shown that twin spaces naturally form a 2-category $\mathcal{T}$ with a monoidal structure, and that operations on twin spaces can be organized into an operad $\mathcal{O}_G$ whose algebras encode a wide range of G -equivariant algebraic structures.

From a conceptual perspective, the main outcome is that twin spaces live at the intersection of:

- group actions and homogeneous spaces $(L \backslash G)$;
- higher category theory (via the 2-category $\mathcal{T}$ and its monoidal structure);
- operad theory (via the operads of twin operations and their algebras).

This viewpoint complements and enriches the deformation cohomology theory of Article A, suggesting that twin spaces should be treated as higher-structured objects whose deformations are naturally approached via 2-categorical and operadic techniques.

Possible directions for future work include:

- (a) **Enriched twin spaces.** Developing twin spaces in enriched categories (e.g. dg-categories, ∞ -categories) so that both $\mathcal{T}$ and $\mathcal{O}_G$ become higher-categorical objects, suitable for derived deformation theory.
- (b) **Homotopical twin operads.** Constructing model category structures on categories of twin operads and their algebras, allowing for homotopy invariance and derived deformation theory in the sense of Lurie and others.
- (c) **Applications to quantum groups and Hopf algebras.** Interpreting quantized structures as deformations of twin operads or as algebras in $\mathcal{T}$ equipped with additional duality and antipode data.

- (d) **Links with higher gauge theory.** Investigating whether twin spaces and their 2-categorical structure can model aspects of higher gauge theories, where fields and symmetries are naturally described by higher groupoids and 2-groups.

These directions suggest that twin spaces are not only a tool for organizing deformations of operations under group actions, but also a promising building block for a broader higher-structural framework at the interface of algebra, geometry, and mathematical physics.

References

- [1] M. Gerstenhaber, On the deformation of rings and algebras, *Ann. of Math. (2)* **79** (1964), 59–103.
- [2] S. Mac Lane, *Categories for the Working Mathematician*, 2nd ed., Graduate Texts in Mathematics, 5, Springer, 1998.
- [3] T. Leinster, *Higher Operads, Higher Categories*, London Math. Soc. Lecture Note Ser., 298, Cambridge Univ. Press, 2004.
- [4] G. M. Kelly, *Basic Concepts of Enriched Category Theory*, London Math. Soc. Lecture Note Ser., 64, Cambridge Univ. Press, 1982.
- [5] J. P. May, *The Geometry of Iterated Loop Spaces*, Lecture Notes in Mathematics, 271, Springer, 1972.
- [6] J.-L. Loday and B. Vallette, *Algebraic Operads*, Grundlehren der Mathematischen Wissenschaften, 346, Springer, 2012.
- [7] C. A. Weibel, *An Introduction to Homological Algebra*, Cambridge Studies in Advanced Mathematics, 38, Cambridge Univ. Press, 1994.